HIRA IRSHAD

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SUMMARY

Computer Science undergraduate (B.Tech CSE + IIT Madras BS Data Science, GATE CS 2026 qualified) who builds and ships AI-powered full-stack products. Completed an AI/ML internship at IIT Roorkee building a Next.js + Node.js + MongoDB platform that uses Gemini (Vertex AI) to turn construction documents into structured data. Has shipped LLM-powered apps with React, Flask and the Gemini API and builds AI-natively with Claude Code.

EDUCATION

Bachelor of Technology (B.Tech), Computer Science and Engineering **Aug 2023 – Jun 2027**
Mahatma Jyotiba Phule Rohilkhand University (MJPRU), Bareilly, UP | CGPA: 8.27 / 10

Bachelor of Science (BS), Data Science and Applications **May 2023 – Present**
Indian Institute of Technology (IIT) Madras, Chennai

EXPERIENCE

1. AI/ML Intern **Sep 2026 – Oct 2026**
Built Environment Laboratory, Dept. of Architecture and Planning, IIT Roorkee | Google Centre for Climate Technology

- Worked on a web platform that converts construction Bills of Quantities (Excel, IFC and PDF) into digital material passports with embodied-carbon estimates, using Gemini (Vertex AI) to extract data from documents.
- Enhanced the frontend and integrated it with the backend APIs, and worked on the database layer (data models, seeding and queries).
- Designed and built a separate module end to end (database, APIs, admin tools and UI pages) and integrated it into the main website.

Tech Stack: *Next.js, React, TypeScript, Tailwind CSS, Node.js, Express, MongoDB, Python, Gemini (Vertex AI), Google Drive API*

2. Web Developer, WebOps Team | Part-time **Nov 2025 – Present**
Saranda House, IIT Madras BS Degree Program | Remote | saranda.iitmbs.org

- Develop and maintain the official house website used by students across the IIT Madras BS program, building new pages and reusable UI components and fixing layout issues.
- Collaborate with the design and content teams through GitHub pull requests and code reviews to ship updates quickly and keep the site consistent.

Tech Stack: *Next.js, React, Tailwind CSS, Sanity CMS, Git/GitHub*

3. Python Developer Intern **Sep 2025 – Oct 2025**
TechSolid Development | Remote

- Built an Expense Tracker application in Python to add, edit, delete and categorize daily expenses.
- Added monthly and category-wise summaries so users can see where their money goes, with data stored in a local database.
- Wrote clean, modular code with input validation and error handling, and presented the project to the mentor at the end of the internship.

Tech Stack: *Python, SQLite*

PROJECTS

SmartSearch PDF: AI-Grounded PDF Topic Search | GitHub **Personal Project**
Python, Flask, PyMuPDF, React 18 (Vite), Gemini API

- Full-stack app that finds every occurrence of a word in any PDF with its page, chapter and surrounding context, and on request asks Gemini to explain what the word means in that exact passage.
- Detects chapters from the PDF's table of contents with a font-size heading fallback; calls the LLM only for rows the user shortlists, keeping search instant and API usage low.

AMP-GEN: AI Document-to-Structured-Data Pipeline | GitHub | Live demo **Project**
Python, Gemini API, Pandas, Seaborn, Streamlit

- Built an LLM pipeline that reads scanned PDF documents with Gemini (OCR + parsing), extracts tables into clean records and maps them to a standard Excel template with JSON export.
- Tested on a 64-item construction Bill of Quantities; added data visualizations and deployed a Streamlit app where anyone can upload a PDF and run the full pipeline.

Learner Success Prediction System: ML & Explainable AI **Group Project**
Python, XGBoost, Scikit-learn, SHAP, Flask, React.js

- Built an end-to-end ML pipeline on the HarvardX-MITx MOOC dataset to predict learner certification outcomes, achieving 98.99% classification accuracy with an XGBoost classifier.
- Integrated SHAP-based explainable AI to surface feature-level attributions (e.g. video engagement, forum participation) behind each prediction for non-technical users.
- Deployed the model through a Flask REST API with a React.js dashboard for real-time prediction, confidence scoring and risk assessment.

ShopAI: Smart Shopping Assistant | GitHub **Group Project**
Python, Flask, JavaScript, HTML, CSS

- Natural-language product search that pulls category, budget, brand and features out of queries like "best gaming laptop under 80000" and ranks results with a deal-score engine (discount, rating, sentiment).

StudyDuel: Competitive Study Tracker | GitHub **Personal Project**
React (Vite), React Router, Firebase Auth, Firebase Realtime Database, Firebase Hosting

- Built a study tracker for two partners preparing for GATE and DSA together: daily task checklists with streaks, subject-wise GATE and topic-wise DSA progress, weekly schedules, an exam countdown and a shared resource library.
- Implemented team pairing with a 6-character invite code (max two members) and email login via Firebase Auth.
- Used Firebase Realtime Database listeners for live sync, powering a head-to-head Battle view that compares both users' daily tasks and LeetCode counts, plus quick nudge messages; deployed on Firebase Hosting.

Student Performance Analysis & Recommendation System | GitHub **Group Project**
Python, Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn

- Developed three ML models on survey-based MOOC learner data: dropout prediction (XGBoost, tuned for recall), marks-range classification (Random Forest) and a study-effort recommendation engine (multi-output regression).

Vehicle Parking Web Application | GitHub | Live **Full-Stack Project**
Python, Flask, SQLAlchemy, MySQL, Bootstrap

- Designed a full-stack, real-time parking management system with a three-tier architecture for user bookings, admin oversight and live slot availability.
- Implemented role-based access control for separate admin and user dashboards, with session-based authentication and password hashing.

StudyMate AI: RAG-based Document Q&A System **In Progress**
Python, pypdf, Sentence-Transformers, ChromaDB, Groq API

- Designing a retrieval-augmented generation (RAG) pipeline that answers questions from uploaded study material using sentence-transformer embeddings and a ChromaDB vector store.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, C, C++, SQL, R

AI & LLM: Gemini API, Vertex AI, Groq API, Prompt Engineering, RAG, Sentence-Transformers, ChromaDB, LLM-based OCR & Document Extraction

Machine Learning & Data: Scikit-learn, XGBoost, SHAP, Pandas, NumPy, Matplotlib, Seaborn, NLP, Computer Vision, PyMuPDF, pdfplumber, openpyxl

Frontend: React.js, Next.js, Vite, React Router, Tailwind CSS, HTML5, CSS3, Bootstrap

Backend: Node.js, Express.js, Flask, Jinja2, REST APIs, JWT, Session Auth, Mongoose, SQLAlchemy

Databases & CMS: MongoDB, MySQL, SQLite, Firebase Realtime Database, Sanity (headless CMS)

Cloud & Deployment: Google Cloud (Vertex AI, Cloud Storage), Vercel, Render, Firebase, Streamlit Community Cloud, Google Drive API, Azure AI Services

Tools & AI-Assisted Development: Git, GitHub, Claude Code, Microsoft 365 Copilot, VS Code, Jupyter Notebook, Ubuntu

CERTIFICATIONS

- Google AI: Gemini Certified Student – Google for Education
- Develop Computer Vision Solutions in Azure – Microsoft
- AI Concepts (AI Fluency) – Microsoft
- Explore AI Basics and Generative AI – Microsoft
- Write Effective Prompts to Achieve Optimal Results – Microsoft
- Work Smarter with Microsoft 365 Copilot Chat – Microsoft
- Data Science & Analytics – HP LIFE
- Kaggle Community Competitor and Python Coder badge – Kaggle
- **TCS iON Career Edge: Certified Young Professional** – TCS iON, Cert ID 119854-27530153-1016

ACHIEVEMENTS & HACKATHONS

- **GATE 2026 (Computer Science & IT):** Qualified in third year of B.Tech.
- **GCI World 2026:** Selected for the data science programme of the Matsuo-Iwasawa Lab, University of Tokyo (Sep 2026).
- **Certificate of Appreciation:** IIT Madras BS in Data Science Programme, for WebOps contributions to Saranda House; Certificate of Excellence, Sundarbans House, IITM BS.
- **Hackathons & contests:** Women Who Master Hackathon, Logitech × Aspire For Her (Jul 2026); Bytequest 1.0 coding contest, Buildon IND (May 2026); TATA Crucible Campus Quiz (2025); Ideathon 2.0, RIF (2024).

WORKSHOPS & TRAINING

- India-AI Impact workshop on generative AI and AI ethics, NIELIT under MeitY, Govt. of India (2026) · AI & Innovation Sprints: Rapid Prototyping for Digital Transformation, Institution of Engineers (India) (Jan 2026)
- Smart Science: AI-Powered Innovation, India-AI Impact Summit 2026 pre-summit event, MJPRU (Dec 2025) · Tracing the Invisible: OSINT & Digital Footprint, IIT Madras BS × InnovXus (Aug 2025)